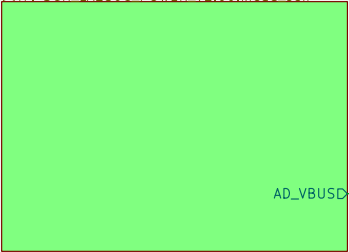
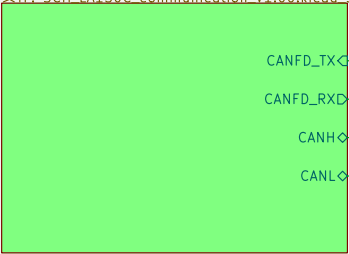


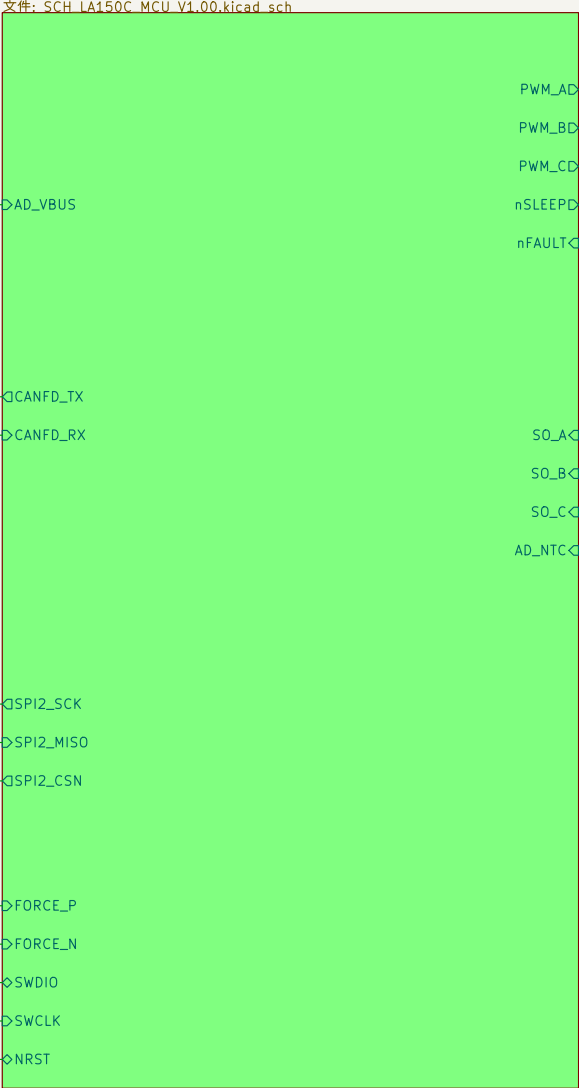
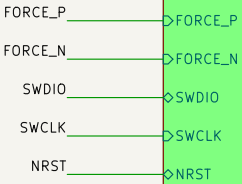
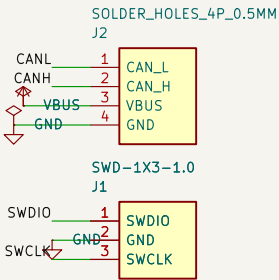
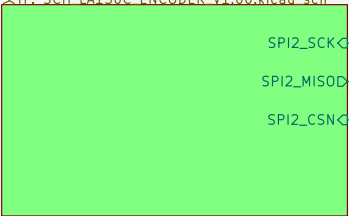
LA150C RDIVER V1.00 – TOP
层次页: POWER / MCU / MOTOR / ENCODER / COMMUNICATION
电源流向: J2 PIN3 VBUS(4.2-15V, 12V 额定) → [D3 SMF16CA] → [SCT2230MLUAR 同步降压] → PWR_3V3 → [MP6543H V3P3 + 磁珠] → PWR_3V3A
微控制器接口: J2 4×0.5mm 焊接口 (0.8mm 外径, 2.54mm 间距) PIN1 CAN_MCU / PIN2 CAN_H / PIN3 12V / PIN4 GND; U/V/W 与 SWD 不引出到该连接器
文件: SCH_LA150C_POWER_V1.00.kicad_sch



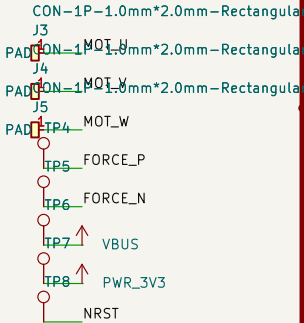
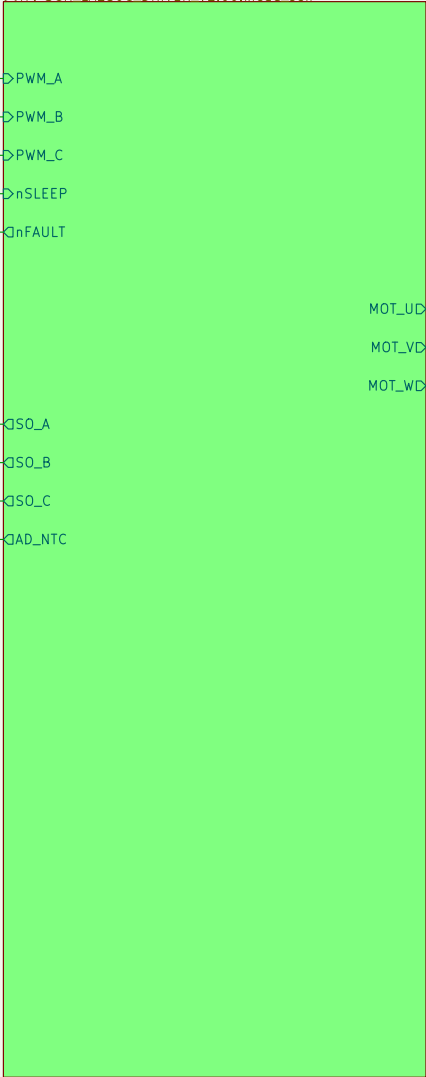
communication
文件: SCH_LA150C_communication_V1.00.kicad_sch



ENCODER
文件: SCH_LA150C_ENCODER_V1.00.kicad_sch



MOTOR
文件: SCH_LA150C_DRIVER_V1.00.kicad_sch



J3/J4/J5 = MOT_U/V/W 焊接口 (1.0×2.0mm Rectangular) : TP4/5 = FORCE_P/N: TP6 = VBUS 4.2-15V: TP7 = PWR_3V3 3.3V: TP8 = NRST

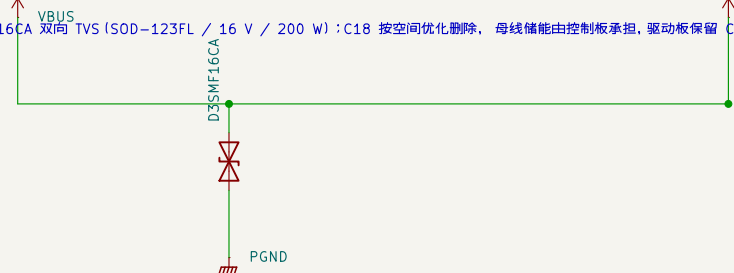
Rev	Date	Author	Changes
V1.00	2026-10-11	Codex	V1.12资源表/EN上拉/外置晶振

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LA150C RDIVER V1.00			
LA150C 集成驱动器原理图			
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LA150C RDIVER V1.00 – POWER
母线 VBUS_IN(4.2-15V) → SMF16CA / 10μF → VBUS → SCT2230MLUAR 同步降压 → PWR_3V3

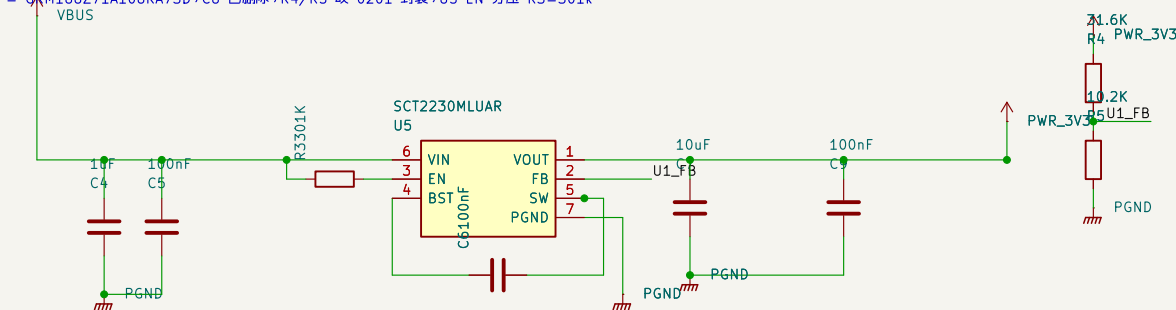
A 输入保护与母线电容

D3 SMF16CA 双向 TVS (SOD-123FL / 16 V / 200 W) :C18 按空间优化删除, 母线储能由控制板承担, 驱动板保留 C4 1μF/50V 本地旁路。



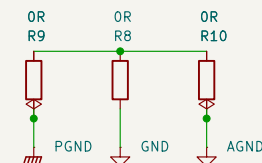
B 同步降压 3.3V – U5 SCT2230MLUAR

U5 按 datasheet Figure 8 (12V→3.3V/2A) 配置:R3 301k 使能分压、C6 100nF BST-SW、R4 31.6k / R5 10.2k 反馈、C7 10μF X7R + C9 100nF 输出 (60mA/200mV 目标)
C7/C21 = GRM188Z71A106KA73D:C8 已删除:R4/R5 改 0201 封装:U5 EN 分压 R3=301k



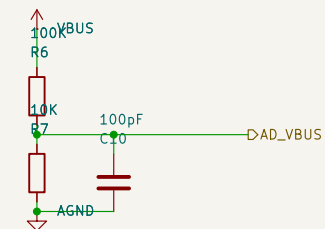
D 地网络星形连接

PGND / AGND / GND 各经 0Ω 在单点汇合
(芯片地星形连接, 避免大电流回流污染采样参考)



C 母线电压检测 → AD_VBUS

分压比 1/11:8V→0.727V, 14V→1.273V, 16V→1.455V:RC 由 100k||10k × 100pF ≈ 0.91μs



Title
LA150C RDIVER V1.00
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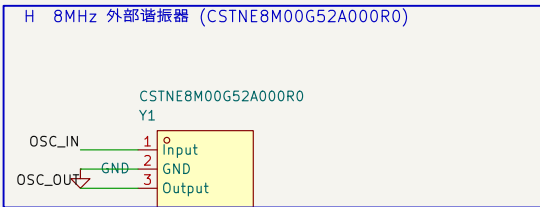
Date

2026-10-04

Rev

V1.00

A4

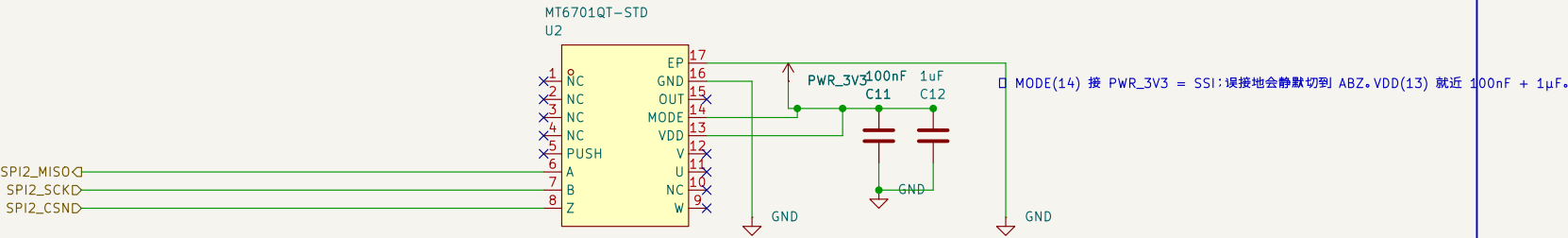


SWD (PA13/PA14 + NRST) 是 $\phi 13$ 内唯一调试/烧录通道, 必须在 PCB 上保留可接入焊盘: SWD 排针放在 TOP 面, NRST: R33 10K 上拉 + C36 100nF; NRST 同时是 Bootloader 上拉 + R37

H 8MHz 外部谐振器 (CSTNE8M00G52A000R0)

时钟采用外部驱动器同款 CSTNE8M00G52A000R0: 8MHz、内置 10pF、SMD3213-3P; pin1→OSCIN(E1), pin3→OSCOU(F1), pin2→GND.

E 磁编码器 MT6701QT–STD (SSI)



SSI:CSN 拉低 → 24 个 CLK → 14bit 角度 + 4bit 磁场状态 + 6bit CRC (X⁶+X+1, MSB first) :CLK 上升沿更新 DO。

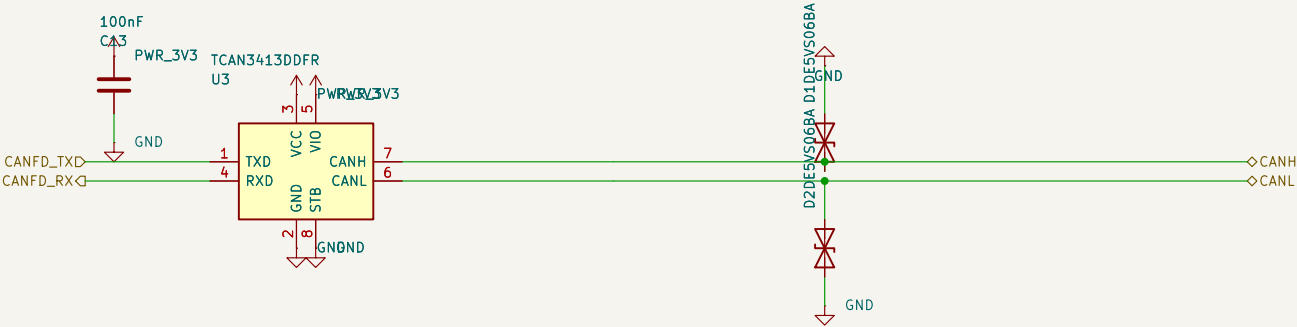
MT6701QT–STD 引脚校正:9=W、11=U、12=V:SSI 模式仅使用 6/7/8/13/14/16/17。

磁体 Ø6×2.5mm N35SH 径向两极, 气隙 0.5-2.0mm (典型 1.0mm) , 偏心 ≤0.3mm (结构件, 不上图)

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LA150C RDIVER V1.00			
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F CAN FD 收发器 TCAN3413DDFR

STB(8) 接 GND = 正常工作模式;高电平为待机。



因板空间受限, 本版取消 CMC 与板内 60.2Ω/4.7nF 分裂终端;总线段终端由线缆/系统端实现。D1/D2 靠近 J2 接口放置。

Title			
LA150C RDIVER V1.00			
LA150C 集成驱动器原理图			
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A4	6/6	2026-10-04	V1.00